TripMate AI: Intelligent Group Activity Planning Platform

Powered by a Unified Deep-Reasoning Agent

Project Idea, Features & Technical Architecture Proposal for Supervisor

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1. Introduction & Project Idea

1.1. The Problem: The Friction of Group Planning

Organizing activities for 3 to 10+ people (whether an international vacation, weekend road trip, dining out, or casual social hangout) consistently suffers from severe real-world bottlenecks:

- **Exhausting & Time-Consuming Manual Research:** Planners must browse dozens of disparate platforms (blogs, Google Maps, review sites, booking portals, flight aggregators) to compile opening hours, ticket pricing, and transit routes—wasting days or even weeks of tedious manual effort.
- **High Friction in Satisfying Everyone's Tastes:** Manually modifying an itinerary whenever a member expresses dissatisfaction is agonizingly slow and complex; adjusting a single stop triggers a chain reaction across transit routes, timing, and group budget, making it nearly impossible to manually tailor a plan that pleases everyone.
- **Fragmented & Messy Group Chats:** Travel ideas, photo links, and flight screenshots get buried under hundreds of unstructured messages on Zalo, Messenger, or WhatsApp, causing critical information to be lost.
- **Organizer Burnout & Single Point of Failure:** Typically, a single organizer bears the entire cognitive and emotional load of researching, polling members, and making compromises, only to face complaints if the group is unsatisfied.
- **Complex Schedule Overlaps ($\pm\delta$ Tolerance):** Member A is free from the 10th to the 20th (± 1 day), while Member B is available from the 15th to the 25th (± 2 days). Manually determining the optimal intersecting window that maximizes attendance is a painful combinatorial puzzle.
- **Conflicting Preferences & Decision Deadlocks:** When half the group desires active mountain trekking while the other half seeks relaxed beach resorts, the group falls into decision paralysis, often leading to canceled plans.
- **Accommodation & Room Booking Bottlenecks:** Finding and coordinating suitable room configurations (bed layouts, group room blocks, cancelation policies) for large parties across multiple booking sites is cumbersome and prone to booking errors.
- **Group Dining Logistics & Split Debt Friction:** Finding venues that handle large-party seating while catering to diverse dietary constraints (allergens, vegetarian, halal) is difficult, and manually calculating post-trip shared debts creates social awkwardness.

1.2. The Solution: A Synchronized Group Room with Human-in-the-Loop AI

TripMate AI transforms group planning into a streamlined, consensus-driven workflow in **under 15 minutes**:

1. **Collaborative Room Chat (Freeform Brainstorming):** All members join a shared room to discuss rough travel wishes, general preferences (e.g., "a warm getaway in autumn" without requiring strict dates upfront), and personal profile constraints.
2. **One-Click Handoff & Proactive AI Clarification (Q&A Loop):** The organizer submits the room's collective discussion to the **Unified Trip Intelligence Agent**. If any critical parameters are missing or ambiguous (e.g., date flexibility, passport status, specific dietary details, budget limits), the Agent proactively posts targeted follow-up questions directly in the chat room to gather full details

from members.

3. **Autonomous Multi-Source Research:** Once fully informed, the Agent resolves schedule overlaps, checks live places/menus, evaluates visa/legal guidelines, and constructs an optimized itinerary.
4. **Granular Leg-by-Leg Voting:** The plan is presented with **stop-by-stop interactive voting** (UP/DOWN/Alternative Proposal). If a specific stop is rejected, the Agent immediately swaps it without breaking the rest of the schedule.
5. **Group Final Consensus & Debt Settlement:** The entire group conducts a final vote to democratically lock the finalized itinerary once consensus is reached, after which automated algorithms simplify all group debts into minimum peer-to-peer transfers.

Figure 1: The end-to-end collaborative planning workflow

2. Platform Features & Core Functionalities

TripMate AI delivers a comprehensive suite of web features tailored specifically for group collaboration:

2.1. 1. Collaborative Group Room & Profile Onboarding

- **Live Gathering Room:** A shared room where members join via invite link or email, chat freely, brainstorm destinations, and input individual profiles (dietary constraints, budgets, rough availability) without needing locked dates upfront.
- **Interactive Agent Q&A Intake:** When submitted, the Agent actively prompts the room with quick interactive clarifying questions to fill in any missing data before generating plans.
- **6 Tailored Event Types:** Built-in templates for Multi-Day Travel (`TRAVEL`), Dining Out (`DINING`), Cafe Hangouts (`HANGOUT`), Entertainment & Arcade (`ENTERTAINMENT`), Cultural Sightseeing (`SIGHTSEEING`), and Custom Events (`CUSTOM`).
- **Role-Based Permissions (RBAC):** Distinct permissions for **Host/Organizer** (creates event, invites members), **Member** (chats, inputs preferences, votes on stops, and participates in final plan locking vote), and **Viewer** (read-only access).

2.2. 2. AI-Powered Planning & Research Hub

- **1-Click AI Generation:** Instant proposal generation from room chat context using DeepSeek-V4.
- **Destination Profiling:** Deep introductions to destinations (e.g., exploring Da Nang's iconic bridges, Ba Na Hills, cultural heritage, and local specialties).
- **Live Inventory & Price Lookup:** Real-time flight search, hotel comparisons, attraction tickets, and tour packages via Amadeus, Booking.com, and Klook APIs.
- **Group Dining Planner:** Filters restaurants by party size capacity, accommodates allergens/dietary restrictions (Halal, vegan), and suggests family-style banquet menus.
- **International Visa & Legal Briefing:** Automatic passport validity check (≥ 6 months), e-Visa guidance, reputable visa agency recommendations, local cultural etiquette, and scam advisories.

2.3. 3. Democratic Voting & Plan Customization

- **Granular Leg-by-Leg Voting:** Independent voting cards on every stop (Hotel, Lunch, Activity) with instant 1-click alternative proposals.
- **Conflict Resolver:** Automatically proposes balanced compromise alternatives when votes are split or tastes are polarized (e.g., Mountain vs. Beach hybrid schedules).
- **Streaming AI Chat Assistant:** Real-time conversation with the AI via WebSockets/SSE to modify itinerary cards on the fly.
- **Manual Drag-and-Drop Plan Builder:** Autocomplete place search via Google Places, auto-calculating route travel times and hotel rates for manual planners.

2.4. 4. Interactive Map & Smart Utilities

- **Interactive Mapview (Mapbox GL JS):** Visualizes timeline stops, routing paths, transit durations, and categorized map pins.
- **Weather-Aware Packing Checklist:** Auto-generates tailored packing items based on OpenWeatherMap destination forecasts and event types.
- **1-Click PDF Export (WeasyPrint):** Generates a polished PDF booklet containing full daily timelines, map snapshots, hotel reservations, and emergency contacts.
- **Bilingual Interface (i18n):** Seamless instant switching between Vietnamese and English.

2.5. 5. Shared Expense Splitting & Debt Settlement

- **Expense Tracking:** Records pool pre-funding (`ADVANCE`) and real expenses (`PAYMENT`).
- **Flexible Split Types:** Supports equal split (`EQUAL`), exact amounts (`EXACT`), or percentages (`PERCENTAGE`).
- **Automated Debt Settlement:** A greedy matching algorithm computes net balances and outputs the minimum number of peer-to-peer bank transfers needed.

3. The Unified AI Agent Architecture & Detailed Operational Flow

3.1. Single Unified Agent Architecture

Rather than dividing context among multiple fragmented sub-agents, TripMate AI uses a **Single Unified Trip Intelligence Agent** equipped with a comprehensive toolset and powered by the **DeepSeek-V4** model hierarchy:

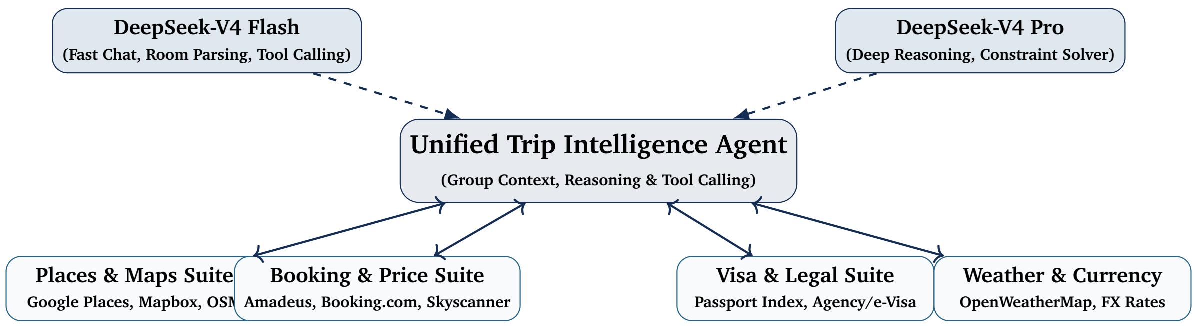

Figure 2: Unified AI Agent tooling ecosystem powered by DeepSeek-V4

- **DeepSeek-V4 Flash:** Handles high-throughput conversational chat, real-time group chat parsing, streaming responses via WebSocket/SSE, and rapid schema validation for API tool execution.
- **DeepSeek-V4 Pro:** Utilizes deep reasoning tokens for complex multidimensional constraint solving, itinerary sequencing under budget limits, and conflict synthesis for divergent member tastes.

3.2. Detailed Step-by-Step Agent Operational Flow

The internal execution lifecycle of the Unified Agent proceeds through six precise sequential phases upon receiving the group's request:

Step 1: **Ingestion, Context Extraction & Proactive Q&A Loop (DeepSeek-V4 Flash):** The Agent parses the chat room stream to extract attendee profiles, general wishes, budget bounds, and date windows $[S_i, E_i] \pm \delta_i$. If essential constraints are missing or ambiguous (such as unstated date ranges, passport/visa status, or dietary specifics), the Agent proactively triggers interactive clarification prompts in the room chat to gather complete information before advancing to heavy computation.

Step 2: **Mathematical Schedule Optimization (DeepSeek-V4 Pro):** Evaluates overlapping availability

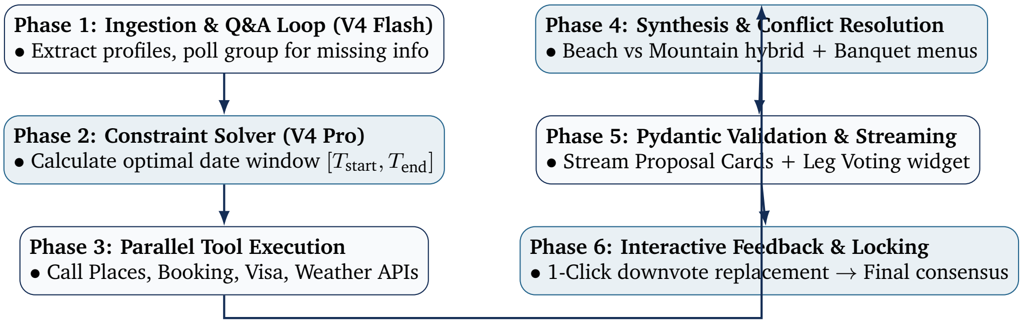

Figure 3: Detailed internal state machine and operational lifecycle of the Unified Agent

intervals to determine the global trip dates $[T_{\text{start}}, T_{\text{end}}]$ maximizing total attendance:

$$\max_{T_{\text{start}}, T_{\text{end}}} \sum_{i=1}^{N} w_i \cdot \mathbb{I}\left([T_{\text{start}}, T_{\text{end}}] \subseteq [S_i - \delta_i, E_i + \delta_i]\right) \tag{1}$$

Step 3: Parallel Multi-API Tool Calling:
- *Place Intelligence:* Queries Google Places/Mapbox for destination profiling, geocoordinates, and transit durations.
- *Live Pricing:* Queries Amadeus, Booking.com, and Klook for flights, hotel room blocks, and entry tickets.
- *Cross-Border Intelligence:* Checks passport validity (≥ 6 months), e-Visa portals, visa agencies, local laws, and cultural taboos.
- *Climate Intelligence:* Fetches OpenWeatherMap forecasts to adapt indoor/outdoor activity allocations.

Step 4: Preference Reconciliation & Itinerary Synthesis (DeepSeek-V4 Pro):
- *Divergent Taste Compromise:* Balances conflicting interests (e.g., hybrid mountain-trekking + beach relaxation schedule).
- *Group Dining & Allergen Management:* Identifies venues with large-party seating and crafts family-style banquet menus compliant with all dietary needs (Halal, vegan, allergen-free).

Step 5: Schema Validation & Streaming Delivery: Formats output into strict **Pydantic v2** JSON schemas and streams real-time Proposal Cards to the frontend via WebSockets / SSE.

Step 6: Granular Voting Loop & Localized Repair: If members downvote a specific leg (e.g., Day 2 Lunch), the Agent immediately invokes a localized tool call to fetch 3 alternative options without recalculating or disrupting the overall itinerary. Once consensus is reached, the plan is locked democratically.

4. Backend Architecture & Data Flow

4.1. Tech Stack & Monorepo Structure

- **Runtime & Framework:** Python 3.11+ with **FastAPI** (high-speed ASGI, native async/await, automated OpenAPI documentation at `/docs`).
- **Database & ORM: PostgreSQL** paired with **SQLAlchemy 2.0 (Async)** and **Alembic** migrations for type-safe database versioning.
- **Caching & Message Broker: Redis** for caching external API lookups (TTL 6–24h), rate-limiting, and WebSocket Pub/Sub synchronization.
- **Background Tasks & Export:** FastAPI BackgroundTasks / Celery for async PDF generation (**WeasyPrint**) and email delivery (aiosmtplib).

4.2. Optimal 3-Table Financial Settlement Engine

Rather than maintaining complex accounting ledgers, the backend computes debts using 3 relational tables (`expenses`, `expense_splits`, `settlements`). A greedy matching algorithm computes net balances ($\text{Net}_u = \sum \text{Paid}_u - \sum \text{Owed}_u$) and produces the minimum number of peer-to-peer bank transfers.

5. Frontend User Experience

Built with **React 18 + Vite + TypeScript**, styled with **Tailwind CSS + shadcn/ui**, and managed via **Zustand** (global state) and **TanStack Query** (caching and optimistic UI updates). The frontend delivers responsive, accessible experiences across mobile, tablet, and desktop viewports.

6. Deployment, DevOps & Security

- **Deployment Pipeline:** Standardized **Docker Compose** for local/staging environments, **Vercel** for Frontend global edge CDN, **Render / Cloud VPS** for Backend Uvicorn workers, and **GitHub Actions** for automated CI testing.
- **Security & Data Protection:** Password hashing using `bcrypt`, dual-token authentication (short-lived JWT Access Tokens + Redis Refresh Token rotation), strict Role-Based Access Control (RBAC), `bleach` input sanitization, and **Pydantic v2** output schema validation.
- **API Rate Limiting & Token Audit:** Redis-backed rate limiting on external and AI endpoints; token usage is recorded in `agent_logs` for complete auditability.

7. Expected Impact & Conclusion

- **15-Minute Planning Velocity:** Replaces days of messy discussions with an AI-driven, consensus-backed itinerary in under 15 minutes.
- **Stress-Free Group Harmony:** Eliminates organizer fatigue, resolves schedule and taste deadlocks mathematically, and ensures fair democratic alignment.
- **Engineering Rigor:** Combines cutting-edge DeepSeek-V4 reasoning, deep API tool integrations, real-time reactive frontend architecture, and robust security standards into an industry-grade software solution.

TripMate AI turns the chaos of group planning into a seamless, collaborative, and intelligent experience.